

CURRICULUM VITAE

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Education

- 2023 **Ph.D.**, Human-Computer Interaction
Carnegie Mellon University
Advisor: Dr. Louis-Philippe Morency
Dissertation: *Multimodal and Dyadic Modeling of Client-Therapist Interaction*
- 2022 **M.S.**, Human-Computer Interaction
Carnegie Mellon University
Advisor: Dr. Louis-Philippe Morency
- 2015 **B.S.**, Computer Science, **B.S.**, Mathematics, *magna cum laude*, 3.6/4.0
North Carolina State University
Minor concentrations: Cognitive Science, Physics
Phi Kappa Phi, Sigma Pi Sigma, University Honors, Computer Science Honors, Mathematics Honors

Employment

- 2024 – 2025 **Postdoctoral Research Fellow**, Robotics Institute
Carnegie Mellon University
The Human Sensing Lab (Dr. Fernando De La Torre, Dr. Lori Holt)
- 2016 – 2023 **Graduate Research Assistant**, Human-Computer Interaction Institute
Carnegie Mellon University
The MultiComp Lab (Dr. Louis-Philippe Morency)
- 2015 – 2016 **Graduate Research Assistant**, Center for Educational Informatics
North Carolina State University
The LearnDialogue Group (Dr. Kristy Boyer), The Intellimedia Group (Dr. James Lester)
- 2013 – 2015 **Undergraduate Researcher**, Center for Educational Informatics
North Carolina State University
The LearnDialogue Group (Dr. Kristy Boyer)

Research Interests

- **SOCIAL SIGNAL PROCESSING:** advanced modeling of social interaction, cognitive-affective modeling frameworks, and multimodal communication analysis for both retrospective and real-time applications.
- **MULTIMODAL MACHINE LEARNING:** integration of verbal and non-verbal modalities, development and implementation of novel probabilistic and causal modeling techniques, and temporal sequence modeling for dynamic data interpretation.
- **AFFECTIVE COMPUTING:** emotion recognition systems, predictive modeling of affective states, and enhancement of human-computer interaction through emotion-adaptive interfaces.
- **ARTIFICIAL INTELLIGENCE:** personalized user modeling and adaptation, comprehensive multimodal behavior analysis, and development of explainable and transparent AI systems.
- **HEALTHCARE & LEARNING TECHNOLOGIES:** AI-driven diagnostic support tools, innovations in computational psychiatry, adaptive intelligent tutoring systems, and the development of personalized learning environments.

Honors and Awards

- Outstanding Paper Award, Twenty-Fourth International Conference on User Modeling, Adaptation, and Personalization, 2016.
- Exemplary Paper Award, Ninth International Conference on Educational Data Mining, 2016.
- Best Paper Award, Twenty-Third International Conference on User Modeling, Adaptation, and Personalization, 2015.
- National Science Foundation Graduate Research Fellowship, 2015.
- North Carolina State University Department of Computer Science Senior Award for Scholarly Achievement, 2015.
- Generation Google Scholarship, 2014.
- North Carolina Society of Information Management Scholarship, 2013, 2014.
- Howard A. Petrea Endowed Scholarship in Mathematics, 2013.

Journal Publications

1. J. Girard*, **A. Vail***, E. Liebenenthal, K. Brown, C. Kilciksiz, L. Pennant, E. Liebson, D. Öngür, L.-P. Morency, J. Baker. Computational Analysis of Spoken Language in Acute Psychosis and Mania. *Schizophrenia Research*, 2021. (*equal contribution)
<https://doi.org/10.1016/j.schres.2021.06.040>

Conference Publications

2. **A. Vail**, J. Girard, L. Bylsma, J. Fournier, H. Swartz, J. Cohn, L.-P. Morency. Representation Learning for Interpersonal and Multimodal Behavior Dynamics: A Multiview Extension of Latent Change Score Models. *Proceedings of the Twenty-Fifth International Conference on Multimodal Interaction (ICMI 2023)*, Paris, France, 2023.
<https://doi.org/10.1145/3577190.3614118>
3. **A. Vail**, J. Girard, L. Bylsma, J. Cohn, J. Fournier, H. Swartz, L.-P. Morency. Toward Causal Understanding of Therapist-Client Relationships: A Study of Language Modality and Social Entrainment. *Proceedings of the Twenty-Fourth International Conference on Multimodal Interaction (ICMI 2022)*, Bangalore, India, 2022.
<https://doi.org/10.1145/3536221.3556616>
4. **A. Vail**, J. Girard, L. Bylsma, J. Cohn, J. Fournier, H. Swartz, L.-P. Morency. Goals, Tasks, and Bonds: Toward the Computational Assessment of Therapist Versus Client Perception of Working Alliance. *Proceedings of the Sixteenth International Conference on Automatic Face and Gesture Recognition (FG 2021)*, Jodhpur, India, 2021.
<https://doi.org/10.1109/fg52635.2021.9667021>
5. **A. Vail**, E. Liebson, J. Baker, L.-P. Morency. Toward Objective, Multifaceted Characterization of Psychotic Disorders: Lexical, Structural, and Disfluency Markers of Spoken Language. *Proceedings of the Twentieth International Conference on Multimodal Interaction (ICMI 2018)*, Boulder, Colorado, 2018.
<https://doi.org/10.1145/3242969.3243020>
6. **A. Vail**, T. Baltrušaitis, L. Pennant, E. Liebson, J. Baker, L.-P. Morency. Visual Attention in Schizophrenia: Eye Contact and Gaze Aversion during Clinical Interactions. *Proceedings of the Seventh International Conference on Affective Computing and Intelligent Interaction (ACII 2017)*, San Antonio, Texas, 2017.
<https://doi.org/10.1109/ACII.2017.8273644>

7. **A. Vail**, J. Grafsgaard, K. Boyer, E. Wiebe, J. Lester. Gender Differences in Facial Expressions of Affect During Learning. *Proceedings of the Twenty-Fourth International Conference on User Modeling, Adaptation, and Personalization (UMAP 2016)*, Halifax, Canada, 2016.
<https://doi.org/10.1145/2930238.2930257>
Outstanding Paper
8. **A. Vail**, J. Wiggins, J. Grafsgaard, K. Boyer, E. Wiebe, J. Lester. The Affective Impact of Tutor Questions: Predicting Frustration and Engagement. *Proceedings of the Ninth International Conference on Educational Data Mining (EDM 2016)*, Raleigh, North Carolina, 2016.
<https://eric.ed.gov/?id=ED592731>
Exemplary Paper
9. W. Min, J. Wiggins, L. Pezzullo, **A. Vail**, K. Boyer, B. Mott, M. Frankosky, E. Wiebe, J. Lester. Predicting Dialogue Acts for Intelligent Virtual Agents with Multimodal Student Interaction Data. *Proceedings of the Ninth International Conference on Educational Data Mining (EDM 2016)*, Raleigh, North Carolina, 2016.
<https://eric.ed.gov/?id=ED592692>
10. **A. Vail**, J. Grafsgaard, K. Boyer, E. Wiebe, J. Lester. Predicting Learning from Student Affective Response to Tutor Questions. *Proceedings of the Thirteenth International Conference on Intelligent Tutoring Systems (ITS 2016)*, Zagreb, Croatia, 2016.
https://doi.org/10.1007/978-3-319-39583-8_15
11. **A. Vail**, K. Boyer, E. Wiebe, J. Lester. The Mars and Venus Effect: The Influence of User Gender on the Effectiveness of Adaptive Task Support. *Proceedings of the Twenty-Third International Conference on User Modeling, Adaptation, and Personalization (UMAP 2015)*, Dublin, Ireland, 2015.
https://doi.org/10.1007/978-3-319-20267-9_22
Best Paper Award
12. **A. Vail**, J. Grafsgaard, J. Wiggins, J. Lester, K. Boyer. Predicting Learning and Engagement in Tutorial Dialogue: A Personality-Based Model. *Proceedings of the Sixteenth ACM International Conference on Multimodal Interaction (ICMI 2014)*, Istanbul, Turkey, 2014.
<https://doi.org/10.1145/2663204.2663276>
13. J. Grafsgaard, J. Wiggins, **A. Vail**, K. Boyer, E. Wiebe, J. Lester. The Additive Value of Multimodal Features for Predicting Engagement, Frustration, and Learning During Tutoring. *Proceedings of the Sixteenth ACM International Conference on Multimodal Interaction (ICMI 2014)*, Istanbul, Turkey, 2014.
<https://doi.org/10.1145/2663204.2663264>
Nominated for Outstanding Paper Award
14. **A. Vail**, K. Boyer. Adapting to Personality Over Time: Examining the Effectiveness of Dialogue Policy Progressions in Task-Oriented Interaction. *Proceedings of the Fifteenth Annual Meeting of the Special Interest Group on Discourse and Dialogue (SIGDIAL 2014)*, Philadelphia, Pennsylvania, 2014.
<https://doi.org/10.3115/v1/W14-4306>
15. **A. Vail**, K. Boyer. Identifying Effective Moves in Tutoring: On the Refinement of Dialogue Act Annotation Schemes. *Proceedings of the Twelfth International Conference on Intelligent Tutoring Systems (ITS 2014)*, Honolulu, Hawaii, 2014.
https://doi.org/10.1007/978-3-319-07221-0_24

Preprints

1. C. Ma*, K. Joo*, **A. Vail***, S. Bhattacharya, Á. García, K. Baker-Matsuoka, S. Mathew, L. Holt, F. Torre. Multimodal Fusion with LLMs for Engagement Prediction in Natural Conversation. *Published to arXiv in September 2024.* (*equal contribution)
<https://doi.org/10.48550/arXiv.2409.09135>

Academic Service

- ACM Conference on Human Factors in Computing Systems (CHI).
- ACM International Conference on Multimodal Interaction (ICMI).
- Association for Computational Linguistics (ACL) Special Interest Group on Discourse and Dialogue (SIGDIAL).
- IEEE Access (Access).
- IEEE Transactions on Affective Computing (TAFFC).
- IEEE Transactions on Learning Technologies (TLT).
- International Conference on Affective Computing & Intelligent Interaction (ACII).
- International Conference on Educational Data Mining (EDM).
- International Conference on Intelligent Tutoring Systems (ITS).
- International Conference on User Modeling, Adaptation, and Personalization (UMAP).
- International Journal of Human-Computer Interaction (IJHCI).
- Journal of Mental Health (CJMH).
- Psychology and Psychotherapy: Theory, Research and Practice (PAPT).
- Public Library of Science (PLOS ONE).

Teaching Experience

2021	Graduate Teaching Assistant , Carnegie Mellon University Human-Computer Interaction Institute
	Spring 2021 05-610: User-Centered Research and Evaluation Dr. Raelin Musuraca and Dr. Motahhare Eslami ~90 students
2019	Graduate Teaching Assistant , Carnegie Mellon University Language Technologies Institute
	Spring 2019 11-776: Multimodal Affective Computing Dr. Louis-Philippe Morency ~30 students
2013 – 2014	Teaching Assistant , North Carolina State University Department of Computer Science
	Fall 2013 CSC 226: Discrete Mathematics for Computer Scientists Dr. Tiffany Barnes ~250 students
	Spring 2014 CSC 230: C and Software Tools Dr. Sarah Heckman ~120 students
2012 – 2013	Mathematics & Physics Tutor , North Carolina State University College Reading & Learning Association: Level II Advanced Tutor Certification

Relevant Coursework

Graduate Coursework

Spoken Dialogue Systems	Artificial Intelligence I, II
Computational Applied Logic	Combinatorics I
Software Engineering	Graph Theory
Numerical Analysis I	Automated Learning and Data Analysis
Reasoning Under Uncertainty	Human Communication and Multimodal Computation
Process and Theory in HCI	Advanced Multimodal Machine Learning
Computer Science Perspectives in HCI	Computer Vision
Social Perspectives in HCI	Data Visualization
Design Perspectives in HCI	Experimental Design for Behavioral and Social Sciences
Cognitive Science Perspectives in HCI	
Interaction Design Overview	

Undergraduate Coursework

Computer Organization and Assembly	Foundations of Advanced Mathematics
Language for Computer Scientists	Calculus I, II, III
C and Software Tools	Applied Differential Equations I
Concepts and Facilities of Operating Systems for Computer Scientists	Introduction to Linear Algebra
Data Structures	Introduction to Modern Algebra for Mathematicians
Automata, Grammars, and Computability	Introduction to Combinatorics
Ethics in Computing	Probability and Statistics for Engineers
Introduction to Artificial Intelligence	Mathematical Analysis I, II
	Mathematics of Scientific Computing
Technology & American Culture	Symbolic Logic
The Creative Process in Science	
Self, Schooling, & Social Order	Introduction to Language & Linguistics
Introduction to College Tutoring	Introduction to Cognitive Science
	Introduction to Psychology
	Cognitive Processes

Professional Membership

- Association for Computing Machinery (ACM).
- Association for the Advancement of Affective Computing (AAAC).
- International Artificial Intelligence in Education Society (IAIED).
- Institute of Electrical and Electronics Engineers (IEEE).
- International Educational Data Mining Society (IEDMS).
- Students & Technology in Academia, Research, and Service (STARS) Student Leadership Corps.